



Analysis of Munich Neighbourhoods for New Immigrants Using Machine Learning

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1. Introduction

1.1. Background

I moved to Germany in January 2026 as a new immigrant. After four months of searching for opportunities in a small city with limited prospects, I made a decision to move to another city. But before packing my bags, I did what any data analyst would do. I built an analysis that helped me make a data-driven decision to move to Munich.

Germany recorded over 262,000 net immigrants in 2024, with Munich consistently ranking as one of the top destination cities for newcomers. [1] A 2025 Mercer Quality of Living Survey ranked Munich as the 3rd best city in the world to live in [2] so it is no surprise that thousands of people from across the globe choose to settle here every year.

Munich has 25 Stadtbezirke (city districts), and they are not all equal. Crime rates vary. Unemployment varies. Rent varies dramatically. As a new immigrant and data analyst, I decided to stop guessing and start analysing.

This project was inspired by Jessica Ayodele's analysis of Toronto Neighbourhoods. [3]

1.2. Problem Statement

Munich's 25 Stadtbezirke span the entire city from the historic Altstadt at its centre to the quieter outer districts on its edges. As a new immigrant, the vital question to answer is: "What district do I settle in?"

The aim of this project is to group Munich's 25 districts in order of desirability for new immigrants using Machine Learning and Data Visualisation techniques, and to present the findings in an interactive Power BI dashboard.

1.3. Basis

There are several factors to consider when settling down in any location. For this project, I performed my analysis using the following criteria:

- Primary Benchmarks: Unemployment rate and Crime rate per district
- Secondary Benchmark: Average monthly rent for a 1-bedroom apartment per district (EUR)

2. Data

2.1. Data Description

For this project, all datasets were obtained from official German and Munich open data portals:

- Crime Rate Data: Crimes per 1,000 residents per district from the Kriminalitätsatlas Munchen (Munich Crime Atlas) 2023. [4]
- Unemployment Rate Data: Percentage of unemployed residents per district from the Statistisches Amt Munchen, Social Atlas 2023. [5]
- Housing Rental Prices: Average monthly rent for a 1-bedroom apartment per district from Immoscout24 rental market report, 2024. [6]
- District Geographic Coordinates: Latitude and longitude point coordinates for all 25 Munich Stadtbezirke via OpenStreetMap and GeoPandas. [7]

2.2. Data Cleaning

The Python modules used for Data Wrangling and Manipulation were Pandas and GeoPandas. All three datasets were merged on the district_id column to produce one final dataframe. See Fig 2.0.

	district_id	district_name	unemployment_rate	crime_rate	avg_rent_eur
0	1	Altstadt-Lehel	4.2	98.2	2100
1	2	Maxvorstadt	3.8	42.1	1950
2	3	Schwabing-West	3.5	35.4	1850
3	4	Schwabing-Freimann	3.2	28.6	1750
4	5	Au-Haidhausen	4.8	52.3	1800
5	6	Sendling	5.2	38.7	1550
6	7	Sendling-Westpark	4.6	31.2	1500
7	8	Schwanthalerhöhe	5.8	45.6	1650
8	9	Neuhausen-Nymphenburg	3.4	29.8	1900
9	10	Moosach	5.6	41.3	1450

Fig 2.0: Pandas Dataframe showing Munich districts and primary benchmarks

3. Methodology

In this section, all Python libraries and techniques used in this project are discussed:

- Pandas - For storing and manipulating structured data. Pandas functionality is built on NumPy
- NumPy - For multi-dimensional array and mathematical operations
- GeoPandas - For storing spatial data coordinates using gpd.points_from_xy()
- Scikit-learn - For Machine Learning tasks: StandardScaler for feature scaling and KMeans for clustering

- Plotly Express - For all interactive visualisations saved as HTML files and hosted on GitHub Pages
- Matplotlib - For early exploratory bar charts during the data analysis phase
- Power BI - For building the final interactive business dashboard with slicers and map

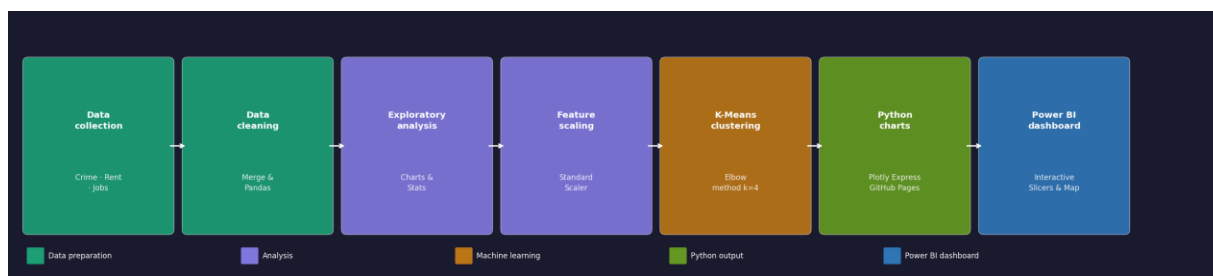


Fig 3.0: Project Flowchart

3.1. Exploratory Data Analysis

3.1.1. Descriptive Statistics for Primary Benchmarks

The average unemployment rate across Munich's 25 districts is 4.7%. The average crime rate is 38.4 crimes per 1,000 residents. The most affordable average rent is EUR 1,180/month (Allach-Untermenzing) while the most expensive is EUR 2,100/month (Altstadt-Lehel). That is a difference of EUR 920/month, nearly EUR 11,000 per year.

3.1.2. Key Observations

- Feldmoching-Hasenberg has the highest unemployment rate at 7.1% - more than double that of Bogenhausen at 2.8%
- Altstadt-Lehel has the highest crime rate at 98.2 per 1,000 residents - driven primarily by city centre activity and tourism
- Rent in Altstadt-Lehel at EUR 2,100/month is 78% more expensive than Allach-Untermenzing at EUR 1,180/month

3.2. Machine Learning Algorithm

A clustering algorithm called K-Means was used to group the districts in order of desirability. K-Means is an unsupervised Machine Learning algorithm. The algorithm works in the following steps:

- i) Determine most optimal k (number of clusters)
- ii) Initialize k such that initial means are randomly generated within the data domain
- iii) k clusters are created by associating every observation with the nearest mean
- iv) The centroid of each of the k clusters becomes the new mean
- v) Steps iii and iv are repeated until convergence is reached

3.2.1. Determining Optimum Number of Clusters

The optimal k was obtained using the Elbow Method. The optimum number of clusters obtained was k = 4.

3.2.2. Feature Scaling

Before clustering, StandardScaler from scikit-learn was used to normalise all three variables. This is a critical step - without it, rent would dominate the clustering over unemployment simply due to the difference in scale.

3.2.3. Clustering Districts by Primary and Secondary Benchmarks

The districts were grouped using all three indicators. The cluster results are shown in Table 1.0.

Cluster	Avg Unemployment	Avg Crime Rate	Avg Rent (EUR)	Label
0	5.71%	42.81	1,395	Least Desirable
1	3.63%	34.89	1,864	Desirable
2	4.20%	98.20	2,100	Semi-Desirable
3	4.02%	27.38	1,502	Most Desirable

Table 1.0: Result of Clustering using Primary and Secondary Benchmarks

4. Results and Discussion

Using the steps explained in Section 3, all 25 Munich districts were ranked into four desirability categories.

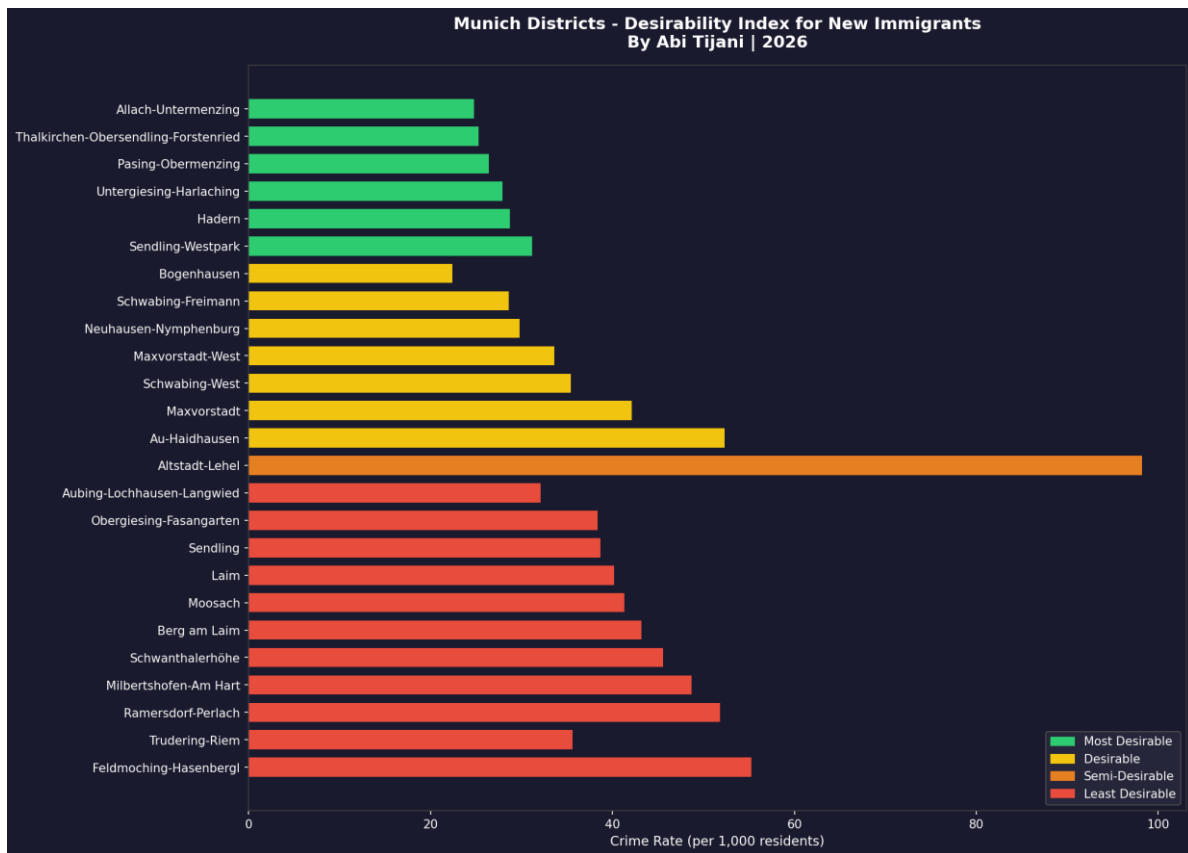


Fig 4.0: Munich Districts Desirability Index

4.1. Most Desirable Districts

Sendling-Westpark, Hadern, Untergiesing-Harlaching, Pasing-Obermenzing, Thalkirchen-Obersendling-Forstenried, Allach-Untermenzing

These 6 districts have the lowest crime rates in Munich averaging just 27.4 crimes per 1,000 residents, moderate unemployment averaging 4.0%, and affordable rents averaging EUR 1,502/month.

4.2. Desirable Districts

Bogenhausen, Maxvorstadt, Schwabing-West, Schwabing-Freimann, Neuhausen-Nymphenburg, Au-Haidhausen, Maxvorstadt-West

These 7 districts have the lowest unemployment rates in Munich averaging just 3.6%. The trade-off is higher average rent at EUR 1,864/month.

4.3. Semi-Desirable Districts

Altstadt-Lehel - Munich's historic city centre with the highest crime rate at 98.2 per 1,000 residents and highest rent at EUR 2,100/month.

4.4. Least Desirable Districts

Feldmoching-Hasenberg, Ramersdorf-Perlach, Milbertshofen-Am Hart, Berg am Laim, Schwanthalerhöhe, Moosach, Laim, Sendling, Obergiesing-Fasangarten, Trudering-Riem, Aubing-Lochhausen-Langwied

These 11 districts have the highest unemployment averaging 5.7% and highest crime averaging 42.8 per 1,000. Rents average EUR 1,395/month.

5. Power BI Dashboard

The findings from the Python analysis were brought into Power BI to build a fully interactive business dashboard. The dashboard allows any new immigrant to filter Munich districts by their own budget and priorities to instantly find their best district.

5.1. Dashboard Features

- KPI cards showing average rent, crime rate, unemployment rate and total districts
- Interactive colour-coded bubble map of all 25 Munich districts
- Bar chart showing districts ranked by crime rate with desirability colour coding
- Desirability slicer - filter by Most Desirable, Desirable, Semi-Desirable or Least Desirable
- Rent range slider - filter districts by maximum monthly rent budget
- Unemployment range slider - filter districts by maximum unemployment threshold
- Full data table showing all districts with desirability labels

5.2. Who is this Dashboard For?

- New immigrants moving to Munich
- Expats being relocated by their company
- HR and relocation agencies helping international hires
- Anyone moving to Munich who wants a data-driven view of where to live

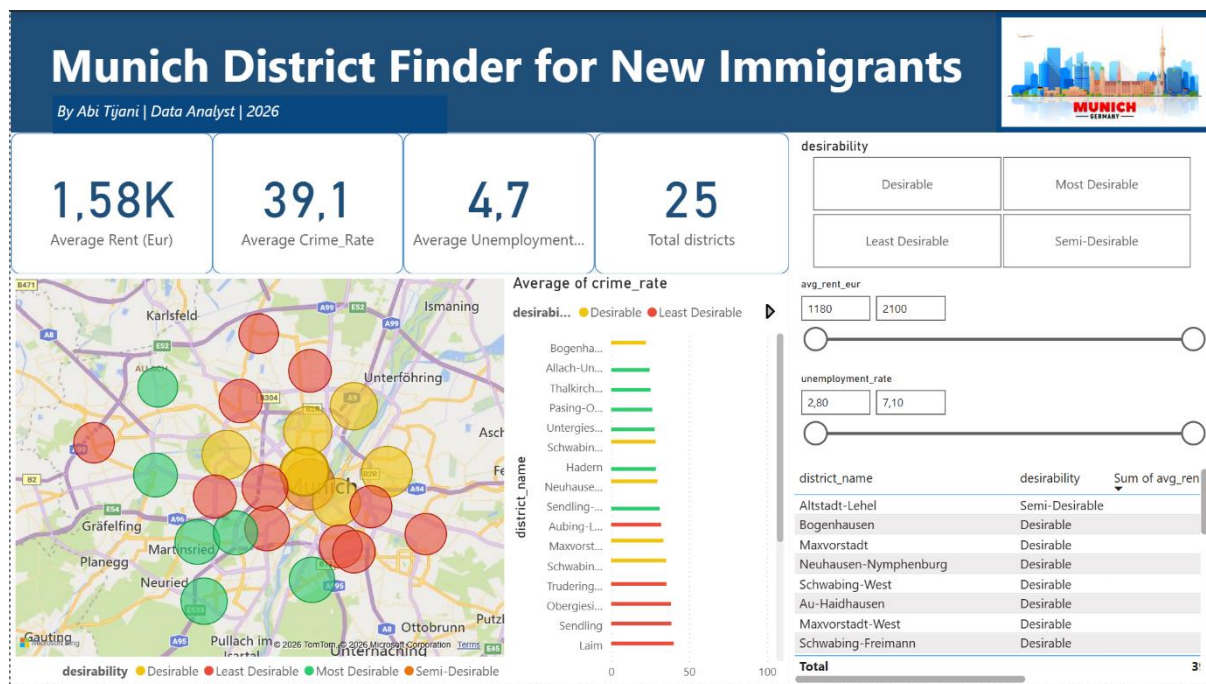


Fig 5.0: Munich Districts Finder PBI Dashboard

Download the full interactive Power BI file (.pbix) from the GitHub repository:
<https://github.com/AbiXData/munich-neighbourhood-analysis>

6. Conclusion

From the results, we can make the following deductions:

- Only 24% of Munich's districts fall into the Most Desirable category, making those 6 districts genuinely worth targeting as a new immigrant
- Most Desirable Neighbourhoods: Consider Sendling-Westpark, Hadern or Pasing-Obermenzing for the best combination of low crime, affordable rent and decent employment rates
- Looking for a Vibrant Community: Bogenhausen and Neuhausen-Nymphenburg offer the best quality of life with lowest unemployment in Munich

- The City Centre Trap: Altstadt-Lehel is exciting but at EUR 2,100/month with the highest crime rate, new immigrants pay a very high premium
- Avoid if You Can: Feldmoching-HasenbergI shows the weakest combination of all three metrics
- Power BI Dashboard: The interactive dashboard allows any new immigrant to filter districts by their own budget and priorities to find their ideal Munich district instantly

References

- [1] Destatis - German Federal Statistical Office, Net Migration Report 2024: <https://www.destatis.de>
- [2] Mercer Quality of Living City Ranking 2025: <https://www.mercer.com/insights/quality-of-living/>
- [3] Jessica Ayodele - Analysis of Toronto Neighbourhoods using Machine Learning, Towards Data Science, 2020
- [4] Polizeiprasidium Munchen - Kriminalitatsatlas Munchen 2023: <https://www.polizei.bayern.de/muenchen>
- [5] Statistisches Amt Munchen - Munchner Sozialatlas 2023: <https://www.muenchen.de/rathaus/stadtinfos/statistik>
- [6] Immoscout24 - Mietpreise Munchen nach Stadtbezirk 2024: <https://www.immoscout24.de>
- [7] OpenStreetMap contributors: <https://www.openstreetmap.org>
- [8] Medium Article: <https://medium.com/@abixdata/analysis-of-munich-neighbourhoods-for-new-immigrants-using-machine-learning-23a732b05981>
- [9] Google Colab Notebook: <https://github.com/AbiXData/munich-neighbourhood-analysis>
- [10] GitHub Repository: <https://github.com/AbiXData/munich-neighbourhood-analysis>